

ISMC121 ANALYSIS OF SYSTEM AND DESIGN

Final Exam

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Department of Computer Science

Comic Shop Frialde Digital Transformation Initiative

Your Comic Shop Frialde

Group 7

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1 PROJECT DESCRIPTION

The Comic Shop Frialde Digital Transformation Initiative aims to provide a fully integrated web-based application for Your Comic Shop Frialde. The project expects to deliver an e-commerce platform for both staff and customers of Your Comic Shop Frialde. Customers will be able to purchase comic books in a way that they can monitor and track their transactions more accurately and experience transparency through the web application. Staff will be able to efficiently manage inventory and customer transactions.

2 BACKGROUND OF THE PROJECT

This project addresses the inventory management of the current system. Experiencing this tedious process during his part-time work for the business, Jearevh has always thought of ways on how to address this problem. Deleting bought comics from facebook albums with 1000 to 2000 images of comics averagely takes 1 to 2 hours. A sale each month averagely aims to open 12 to 18 facebook albums in a span of 3 to 8 days with a billing period 3 to 5 days after the sale period. This puts staff on the clock, prioritizing the billing process before starting on the deleting process amidst having similar execution of tasks. In addition, facebook can also hinder access because of repetitive actions such as scrolling through multiple albums successively or jumping from one image to another quickly.

Problems:

1. Tedious processes

- a. Manual billing process
- b. Inventory process
- c. Removes unnecessary re-upload process due to bogus buyers
- d. Problems in Facebook management

2. Schedule constraints

- a. Each sale period is scheduled every month where albums should be ready for display.
- b. After the sale period, billing also starts every month which necessitates staff to solely prioritize the billing process.

3 SIGNIFICANCE OF THE PROJECT

The project will smoothen transactions, improve inventory management, and increase customer interaction. The idea is to address present inefficiencies and deliver a more seamless purchasing procedure and efficient inventory management. The proposed system will require less staff for inventory management and billing as the web application innately does these processes. This will also remove the monthly sale constraint as the processes will happen instantaneously, removing the requirements of a deleting and billing period in the business schedule. The system also introduces transparency for customers for each transaction they do with the business. Overall, this web-application reduces the number of manual processes and an opportunity for numerous digital improvements for the business.

4 CURRENT SYSTEM

As explained in the background section, the main processes that this project addresses is the deletion and billing processes. Deleting process spans for around 4 to 10 days depending on the amount of albums required to process and the number of comics that are bought in each album. An album with less sales is processed faster because there are few images to delete. Albums with a large number of sales require much more time because each image has to be deleted one by one. An image with multiple copies of a comic will have a comment displaying how many comic stock of that specific comic are left, and the other comments that indicated previous purchase are deleted. This ensures that for the next deletion process, staff can easily determine how many copies are left and in the billing process, which are the new purchases. Billing process is much faster, noting how many copies a customer buys from an album with a corresponding album price.

4.1 Business Processes

Transaction Processes

This process encompasses the entire lifecycle of a customer's purchase, from browsing products to completing a purchase order.

Inventory Database

This process manages the inventory of comic books, ensuring accurate stock levels and efficient restocking.

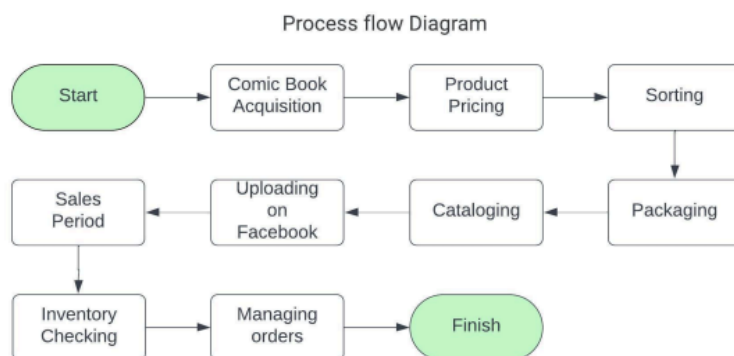
Customer Database

This process involves managing customer information to support personalized service and marketing efforts.

Inventory management

This process involves the preparation of comic books from pricing to packaging, and the management of comic books to be uploaded for sale.

4.2 Process Flow Diagram of the current business processes



The organization has a simple step-by-step procedure for managing comic book transactions from product pricing to managing orders. We identified 7 processes, starting with product pricing. After receiving a batch of new comic books, each book is priced based on its current market value. Each book is then sorted by name

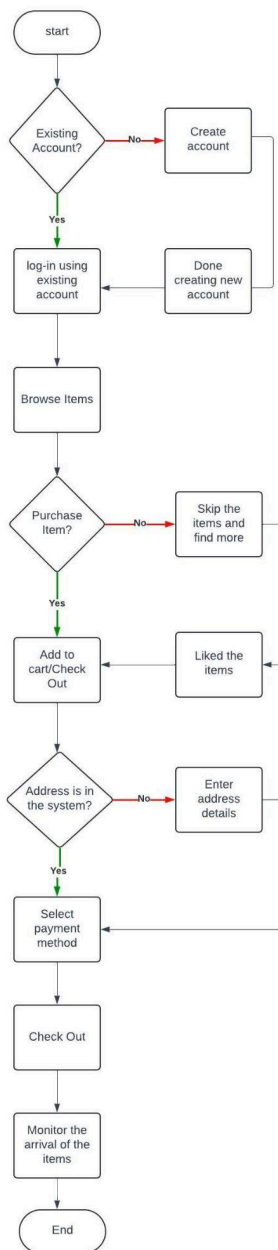
before entering the packaging stage. After packing each book, they are cataloged using a phone. Photos are then uploaded to Facebook albums in the Shop account. The albums are then opened to the public, starting the sales period, which ends after a certain amount of time. After closing the albums, each book in each album is scanned for purchase. Purchases are recorded, and messages to buyers are sent to confirm the order.

5 PROPOSED SYSTEM

The proposed system is a web-based application designed to automate and streamline business processes. It will include modules for automated transactions, real-time inventory management, and comprehensive sales analytics. The system will also maintain a robust customer database and enhance website functionality.

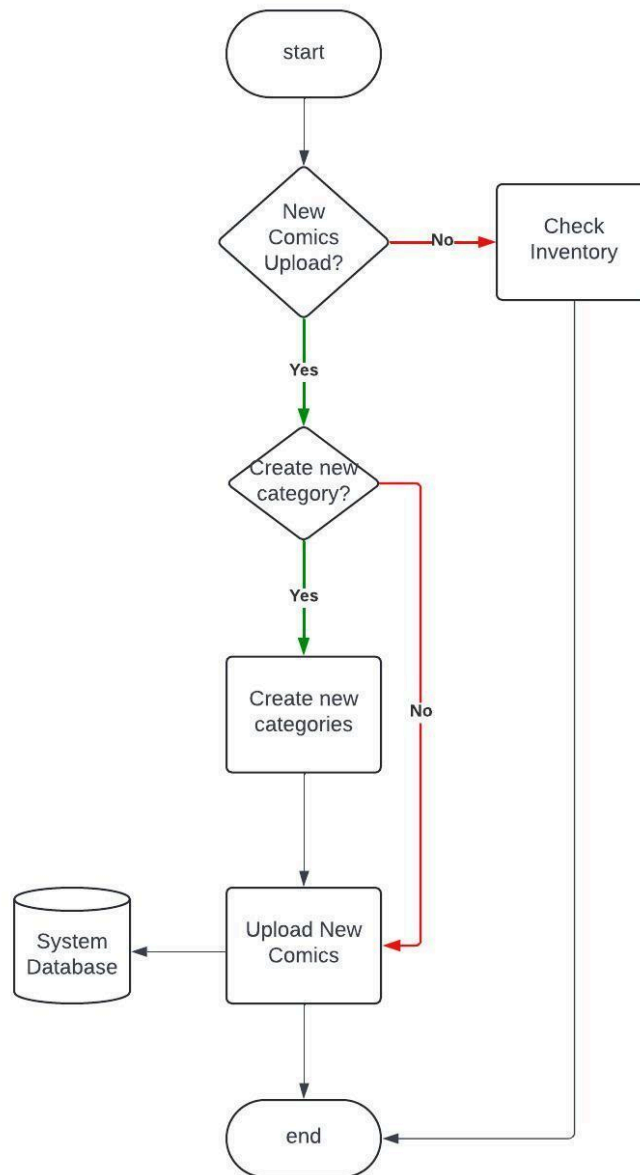
5.1 Process Flow Diagram

Customer Process Flow Diagram



This process flow diagram shows the perspective of the customer in using the system. The system prompts the user to log-in or create an account to access the website. Afterwards, they are directed to the home page where they can browse categories and items. The user can purchase their desired item. In the cart system, the user is required to input their address to complete the checkout process if they haven't done so already during the signup process. The user can track the item's status order.

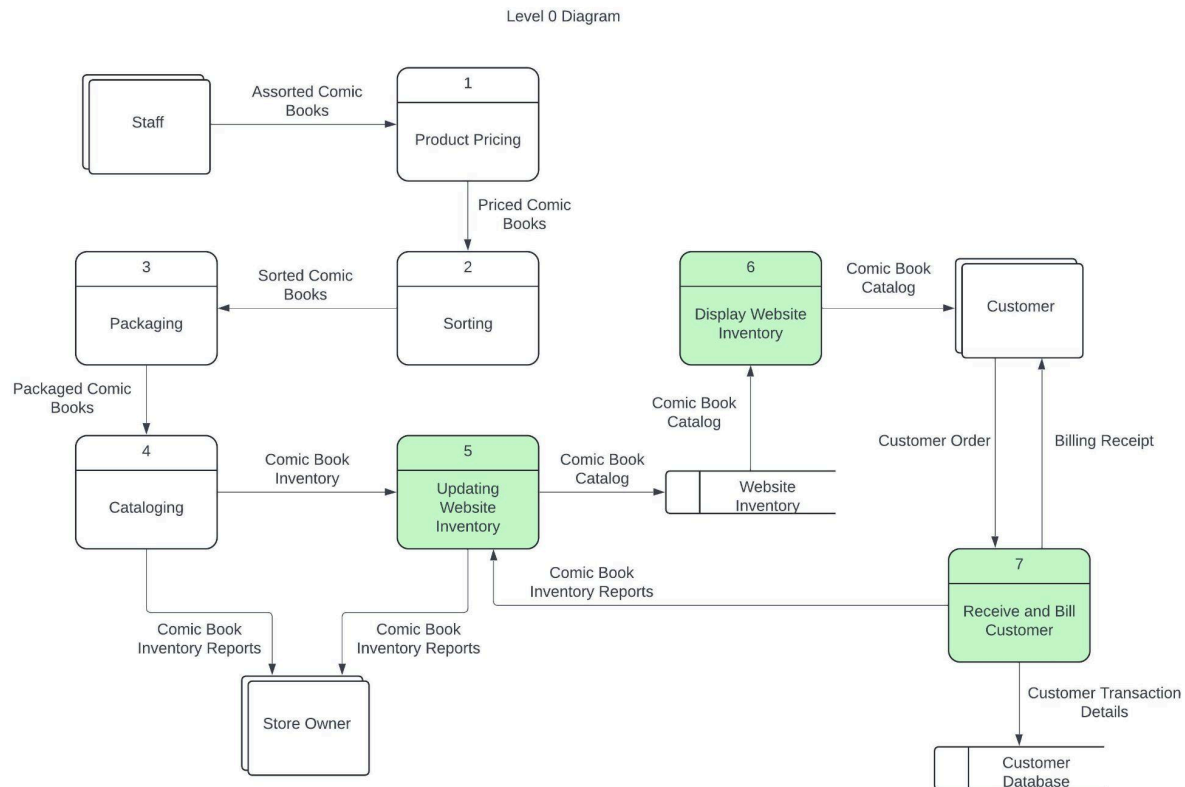
Staff Process Flow Diagram



This process flow diagram shows the perspective of the staff in using the system. The staff can either upload comic books or just check the current inventory. The staff can upload new comic books into existing categories or create a new category. The uploaded comic books are stored in the database.

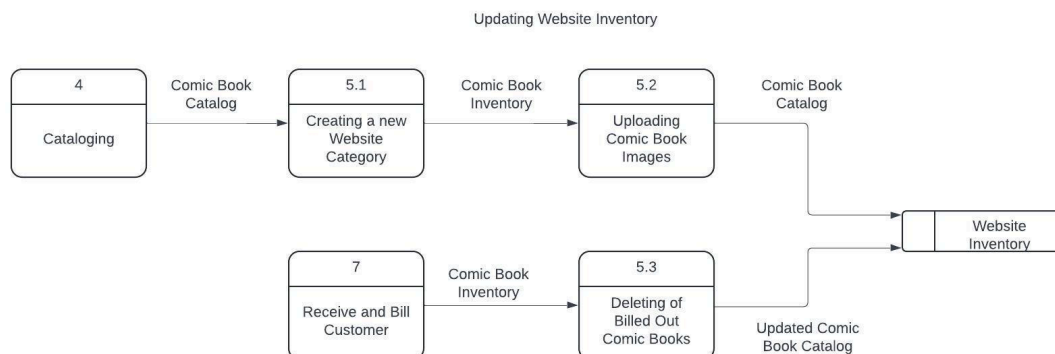
5.2 Data Flow Diagram

Level 0 Diagram



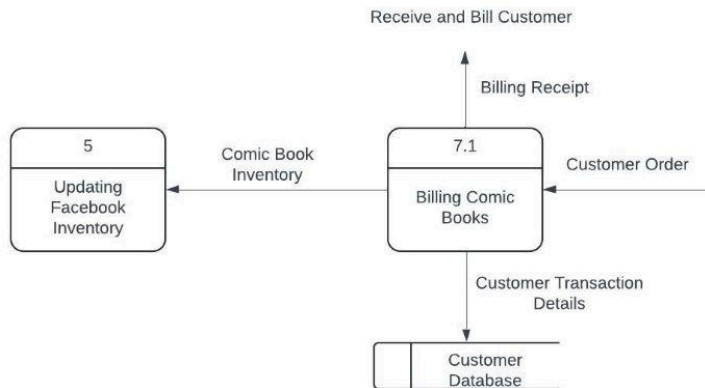
This shows the proposed Level 0 diagram of the system. Processes 1-4, which is the preparation stage, are still the same as the current system. Process 5, which is the manipulation of the inventories, now accesses the website database, and process 6 now displays the website inventory. Process 7 also now happens within the website automatically rather than manually.

Process 5 Explosion Diagram



In process 5, the system allows staff to create categories before uploading the comic book catalog. Also, comic books that have been successfully purchased are deleted from the system.

Process 7 Explosion Diagram



In process 7, the order of the customer goes to the cart. After checking-out the items, the details of purchase goes to the customer database to record customer transaction history.

5.3 CRUD Matrix

Module	Customer	Employee/Staff	Manager
Product	R	C, R, U, D	C, R, U, D
Order	C, R, U, D	R, D	C, R, U, D
Account	C, R, U	R	C, R, U, D
Inventory	-	C, R, U, D	C, R, U, D
Sales	-	C, R, U, D	C, R, U, D

This ensures that all necessary data operations are well-managed across the system's modules for each user role. By detailing the specific operations for Products, Orders, Accounts, Inventory, and Sales, the Comic Shop Frialde Digital Transformation Initiative aims to streamline processes, enhance inventory management, improve customer account management, and provide valuable sales analytics, ultimately leading to a more efficient and transparent business operation.

5.4 Functional and Non-Functional Requirements

Functional Requirements	Non-Functional Requirements
Automated transaction processing	System reliability and uptime
Real-time inventory updates	Data security and privacy
Sales data analytics and reporting	User-friendly interface
Customer database management	Scalability for future growth
Website order management	Fast performance and response times

5.5 User Stories

Transaction Processes

- As a customer, I can add comic books to my shopping cart, so that I can easily purchase multiple items in one transaction.
- In order to track my purchase history as a customer, I can view my past orders.

Sales Analytics

- As a manager, I can view sales reports, so that I can analyze sales trends and make informed decisions.
- As a sales analyst, when reviewing monthly performance, I want to access detailed sales analytics because it helps in identifying successful strategies.

Inventory Database

- As an employee, I can update the inventory database, so that the stock levels reflect the current availability of comic books.
- In order to avoid stockouts as an employee, I can set up alerts for low stock levels.

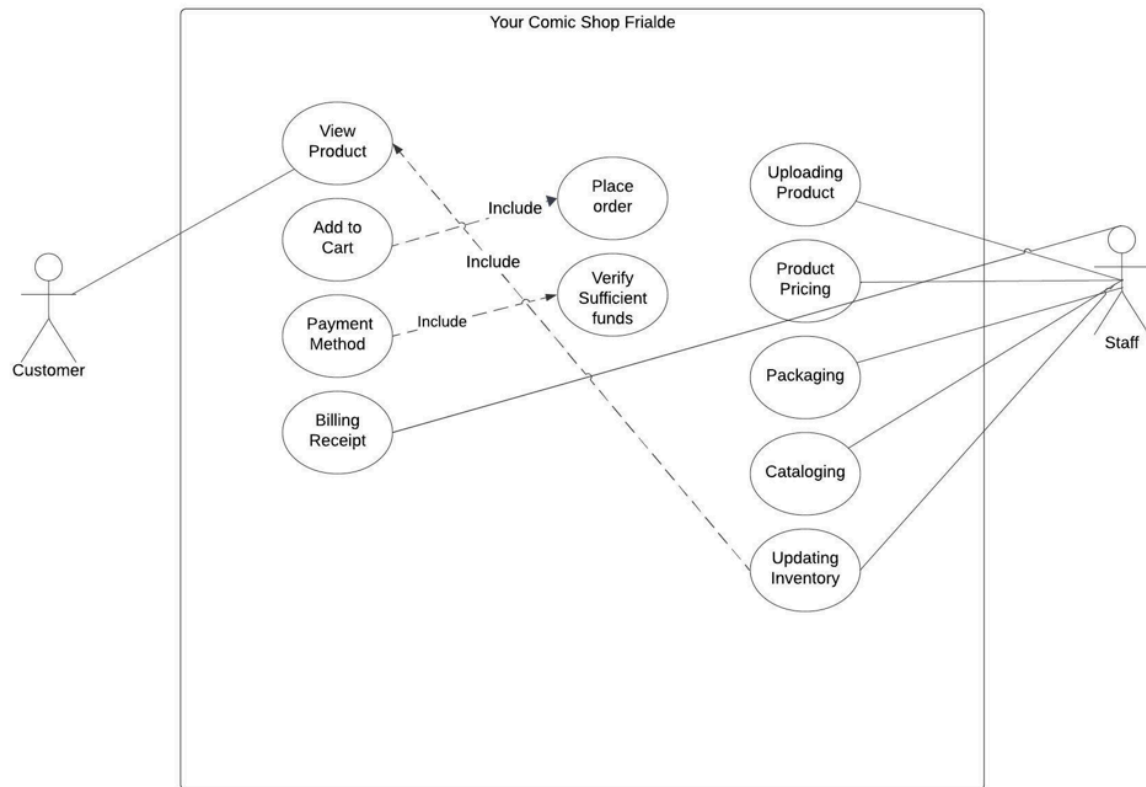
Customer Database

- As a customer, I can update my personal information, so that my contact details and preferences are always current.
- As a customer service representative, when assisting a customer, I want to access their purchase history because it helps resolve issues efficiently.

Website Process

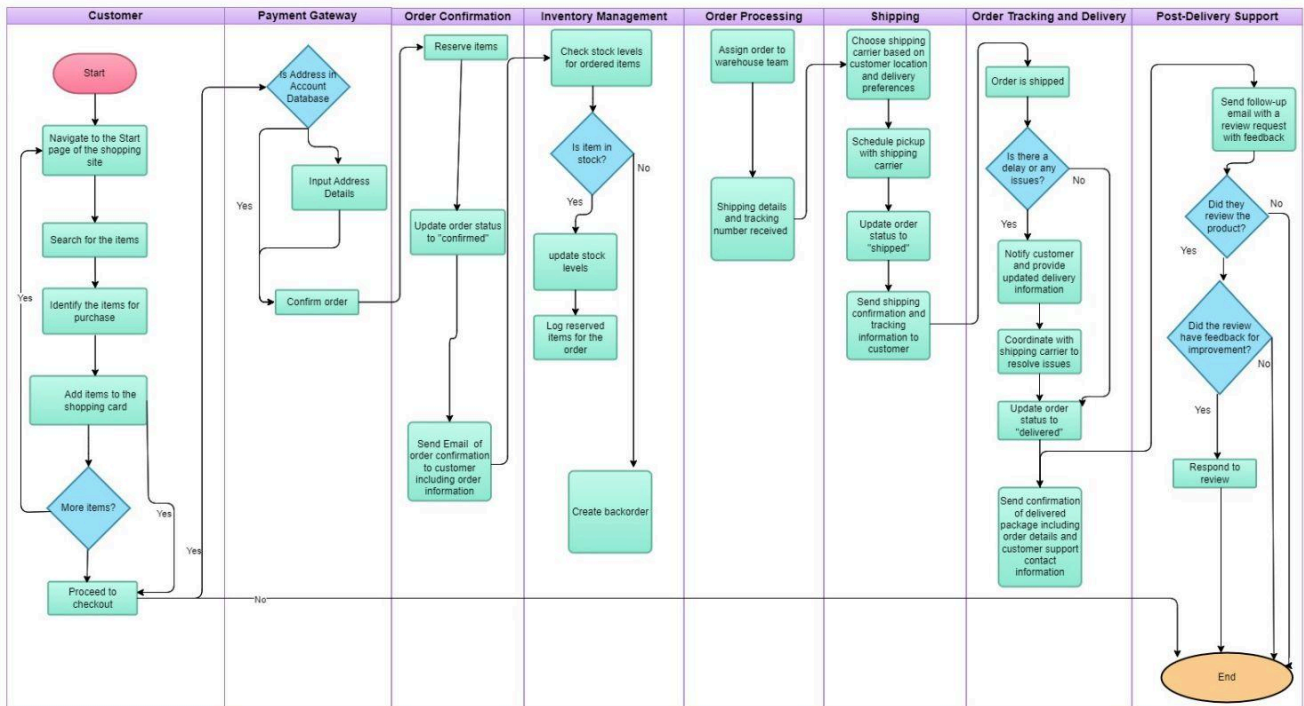
- As a visitor, I can browse comic books by category, so that I can easily find comics that interest me.
- In order to enhance user experience as a web developer, I can implement a search function.

5.6 Use Case Diagram



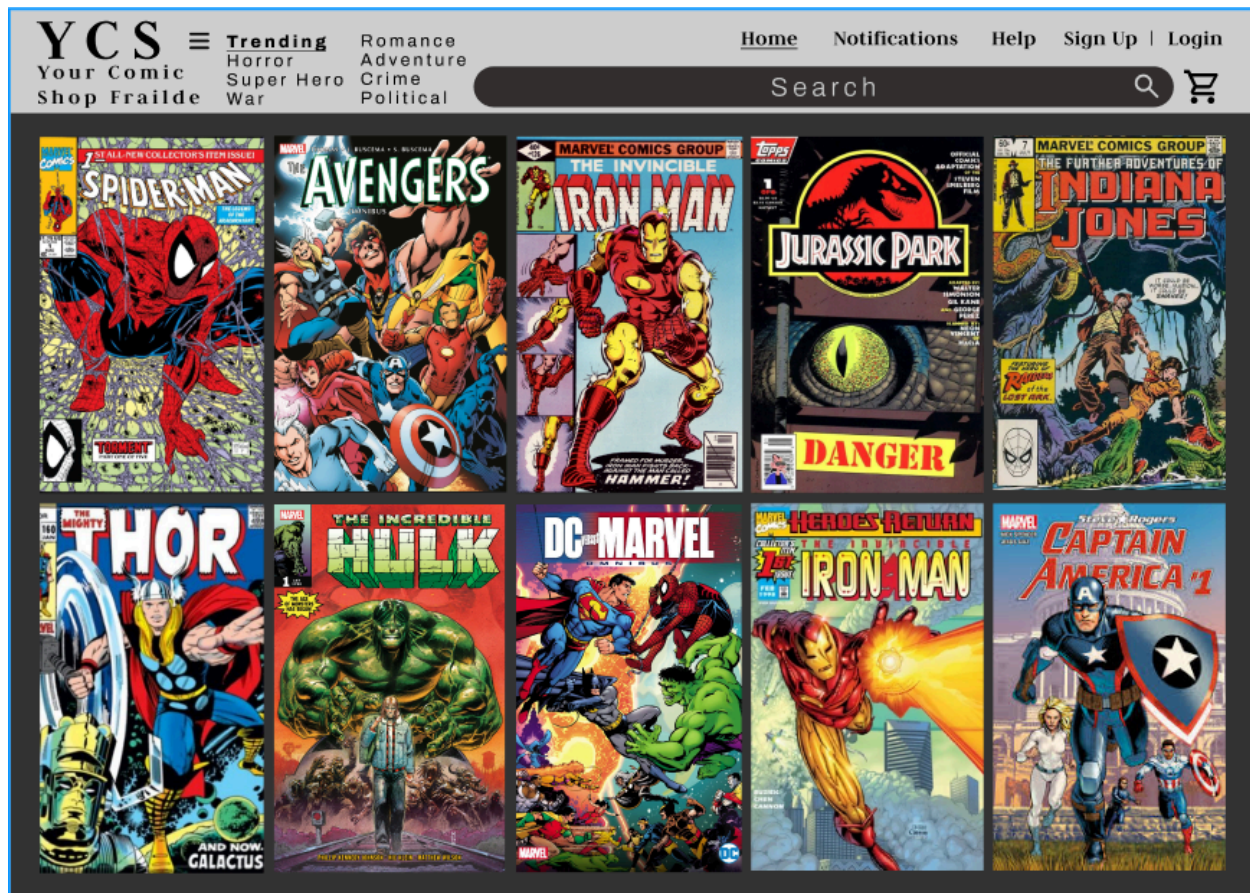
Using the system, the staff can upload and update inventory which are the most important activity in the system. Customers can view the inventory being uploaded and updated on the website. Customers can place orders in the system which are processed by the system and confirmed by the staff.

5.7 Activity Diagram

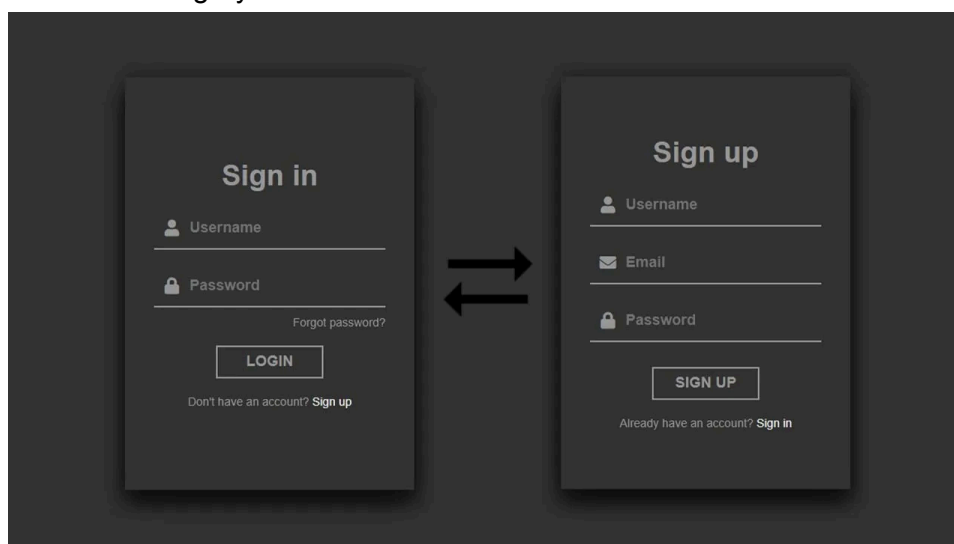


This activity diagram shows the perspective of the customer in using the system. The customer, after logging-in, is brought to the Home page. After browsing for items, the customer can either end their session or proceed to check-out. The customer has to have completed their account details, specifically the address for delivery. The order is received by the system and a confirmation is sent for verification. The staff then checks for the purchased item's availability, if not a backorder is created. The staff then packages the comic books with details from the customer purchase receipt. Then a standard shipping procedure is observed, shipping details reflected in the website through status order. After delivery, the customer may leave a review, when it is a feedback, staff may respond to let the customer know they are heard.

5.8 Sample User Interface



This is the home page that has the navigation bar which contains, home, notifications, help, signup, and login buttons. It also has a search category function that shows different genres. The image takes the user to a specific category page where it displays the comic books inside the category.



This is the sign-in and sign-up design we want to implement that we got from the internet for future development. The following are the design plans for future development:

Homepage:

Header: Logo, navigation menu, search bar, Add to Cart Button

Main Section: Featured comics

Product Page:

Product Image: Display comic book cover

Product Details: Title, author, description, price

Add to Cart Button: For purchasing

Cart Page:

Cart Items: List of selected comics

Total Price: Sum of item prices

Checkout Button: Proceed to payment

Checkout Page:

Billing Information: Input fields for customer details

Payment Information: Input fields for payment details

Confirm Purchase Button: Finalize order

6 APPENDIX

Questionnaire:

Interview Topic: General Flow for Jonale Frialde - Business Owner

Interview Type: Semi-structured interview

- Can you provide an overview of Your Comic Shop Frialde's current business model and operations?
- What are the main challenges or pain points you encounter in running the business?
- How do you currently handle transactions, inventory management, and customer interactions?
- What are your short-term and long-term goals for the business?
- How do you envision the digital transformation initiative impacting your business?

Interview Topic: Operational Flow for Jocelyn Natividad - Staff/Employee

Interview Type: Semi-structured interview

- Can you walk us through the current process of handling transactions and managing inventory?
- What are the common issues or challenges you face during your daily tasks?
- How do you currently communicate with customers and handle inquiries or orders?
- Are there any manual tasks or repetitive processes that you find particularly time-consuming?
- How do you think technology or automation could improve your workflow?

Interview Topic: Customer Experience and Feedback for Customer

Interview Type: Interview Type: Structured interview with rating scale

Rating Questions: Scale of 1 to 4 (1 being the **lowest** and 4 being the **highest**)

- How would you rate your overall shopping experience with Your Comic Shop Frialde?
- How much did factors like product variety, pricing, and customer service influence your decision to purchase from Your Comic Shop Frialde?
- How satisfied were you with the range of comic books available at Your Comic Shop Frialde?
- How would you rate the ease of browsing products and the checkout process on Your Comic Shop Frialde's platform?
- How would you rate the level of customer service provided by Your Comic Shop Frialde?
- Did you encounter any challenges or issues during your shopping experience with Your Comic Shop Frialde? If so, please rate from 1 to 4 in terms of severity.
- How likely are you to recommend Your Comic Shop Frialde to others based on your experience?

Feedback Questions:

- Are there any additional features or services you would like to see implemented to enhance your shopping experience with Your Comic Shop Frialde?
- Is there anything else you would like to share about your experience or suggestions for improvement?